STAT4C03/6C03 Generalized Linear Models

Assignment 1 Due at 10pm on January 29, 2026

Students are reminded that submitted assignments must be entirely their own work. Submission of all or part of someone else's solution (including solutions from the internet or other sources) under your name is academic misconduct and will be dealt with as such. Penalties for academic misconduct can include a 0 for the assignment, an F for the course with an annotation on your transcript and/or dismissal from your program of study.

Instructions:

1. The only way to submit your assignment is through the Crowdmark page for the course. Use the link on Avenue (Content > Crowdmark) to login to Crowdmark and follow the instructions there to upload your assignment.

2. It is your responsibility to ensure that your submitted solutions are readable and complete. Any submitted file which cannot be opened will be given a mark of 0.

3. You may use R Markdown or Knitr to compile the R questions in your assignment but are not required to do so. For any question using R, you are required to include both the R code and output in your solutions as well as your written comments on the model(s) fitted.

4. Late assignments will be automatically assessed a penalty of 2% per hour, or part thereof, that they are late according to the Crowdmark time stamp.

5. You may update your submission up until the deadline. Once the due date has passed, you will no longer be able to update any submission.

6. Please indicate **clearly** on your solutions whether you are in STATS 4C03 or STATS 6C03.

7. You are expected to show all details of your solution and any results taken from my notes or the textbooks must be clearly and properly referenced.

8. Students with accommodations through Student Accessibility Services who require accommodations for this assignment must contact the professor **at least 48 hours before the assignment is due** to request any such accommodations.

9. No other extensions to the due date and time will be given except in extreme circumstances.

10. There are 25 marks for each question so the total mark will be out of 100 for STATS 4C03 students and 125 for STATS 6C03 students.

Q. 1 Consider the data in `Grades.csv` giving marks for homework and three term tests in a statistics class.

a) Fit a model to predict the final score from the homework score and the three term tests. Write a brief report explaining the results of your analysis.

b) The professor wishes to estimate the mean final score for students who get a homework mark of 85 and a mark of 70 in each of the term tests. Give a point estimate and 95% confidence interval for this quantity.

c) Consider a model with 2 covariates – the homework score and the average of the three term tests. Verify that the Wald and likelihood ratio tests of this model relative to the full model in (a) are exactly identical and give the conclusion of the test including the p-value.

d) Using the reduced model from (c), give a point estimate and confidence interval for students with homework score of 85 and an average term test mark of 70 and compare the interval to the one found in part (b).

Q. 2 Products of covariates are often used to introduce interactions into a linear model where the effect of one covariate is different depending on the value or level of another covariate. Suppose that we have a continuous response variable, Y, and two categorical covariates, C_1 and C_2 and let the means of the response variable be as described in the table below. Let level 1 be the reference category for C_1 and level A be the reference category for C_2.

		C_2	
	Level	A	B
C_1	1	μ_{1A}	μ_{1B}
	2	μ_{2A}	μ_{2B}
	3	μ_{3A}	μ_{3B}

a) Suppose we only include the dummy variables for the two covariates in our design matrix, write the means in the table above in terms of the parameters $\beta_0, \beta_1, \beta_2, \beta_3$ of the resulting linear model.

b) Now suppose we also include two further terms in the model corresponding to products of the two dummy variables for C_1 with the one for C_2. Describe the means in terms of the parameters of this new model and compare what you have with the previous result. Give an interpretation of the two new parameters β_4 and β_5.

c) Now suppose that we add a continuous covariate X into the model from part (b). Show that the 6 means above would now be linear functions of the value of the continuous covariate and compare those lines with each other.

d) Explain what happens to the linear relationships between Y and X if we also include the following terms in our model

(i) The product of X with each of the dummy variables for C_1.

(ii) The product of X with all of the dummy variables and their products as used in the model in part (b).

Q. 3 The dataset `Salary.csv` contains data on the salary of 46 computer professionals in a large corporation. The 4 columns in the dataset are as follows:

Name	Description
Salary	Annual salary
Exp	Number of years experience
Educ	Highest education (1=High School, 2=Bachelor's Degree, 3=Advanced degree)
Manage	Management status (1=has management responsibilities, 0=does not)

a) Construct a plot of the Salary against Experience using different colours for the 3 levels of education and different plotting characters for the management status.

b) Fit a linear model which gives a straight line relationship between Salary and Experience with different intercepts and slopes in each of the 6 combinations of education and management status. Write out the 6 fitted lines and add them to the plot from part (a).

c) Fit a reduced model in which the 6 lines are parallel and give the 6 fitted lines.

d) Conduct a test of the hypothesis that the true relationship has 6 parallel lines.

Q. 4 Although generally not a good idea, we can sometimes omit the constant in a linear regression. One instance when we may want to do this is if we have a single categorical covariate with d levels. Instead of the intercept and $d-1$ dummy variables we could include all d dummy variables in the model without the intercept. Suppose that we have n individuals and n_j occurrences of each level $j = 1, \ldots, d$ in our dataset and we decide to take this approach of including d dummy variables and no intercept. Let $\overline{y}_j$ and s_j^2 denote the sample mean and variance of the response variable for individuals in each category $j = 1, \ldots, d$ of the covariate.

a) Describe what the design matrix $\boldsymbol{X}$ looks like in terms of its entries, the sum of each row and the sum of each column.

b) Write down $\boldsymbol{X}^t\boldsymbol{X}$ and, hence, its inverse.

c) Find the estimators $\hat{\beta}_j$ of β_j, $j = 1, \ldots, d$.

d) Give the residuals and hence the mean squared error of the model in terms of the s_j^2, $j = 1, 2, \ldots, d$.

Q. 5 STATS 6C03 Students Only

a) Show that the usual two-sample pooled t-test of equality of two population means can be written as the test of a parameter in a linear model, giving the model and details of the test we would conduct from that model (null hypothesis, test statistic, null distribution etc.). You must prove that the test statistic from the linear model is exactly the same as the test statistic in the usual pooled two-sample t-test.

b) The one-way analysis of variance extends the two-sample pooled procedure to test the equality of $d > 2$ populations. Describe the linear model in this situation, the null hypothesis, test statistic and null distribution we would use in this case. Again show the test statistic and its distribution are exactly the same as in the usual test statistic in a one-way ANOVA table.